

A.A.U CHAMO INTERNATIONAL BUSINESS AGENCY SERVICES LIMITED

Enterprise ERP Dedicated Manual Series

Volume 1: Super Admin & System Administrator Manual

Module 01 Focus: System Governance, Security, Access Control & Executive Dashboard

[!IMPORTANT] CONFIDENTIALITY & SECURITY NOTICE This manual is intended strictly for authorized **Super Administrators** and **System Owners** of A.A.U Chamo International Business Agency Services Limited. It contains step-by-step procedures for managing system security, user permissions, audit chains, and enterprise configurations.

1. Executive Summary & Role Definition

As a **Super Administrator** or **System Administrator** in the AAU Chamo ERP system, you hold full, company-wide operational authority over the platform. Your primary responsibilities include:

- System Governance:** Onboarding users, configuring organizational structures (Business Units and Stations), and assigning role-based permissions.
 - Security & Compliance:** Overseeing user access, enforcing password policies, managing AES-256 data encryption keys, and revoking suspicious active sessions.
 - Audit Chain Integrity:** Verifying the cryptographic integrity of the system's tamper-proof audit trail.
 - Operational Oversight:** Monitoring real-time company KPIs, multi-station revenue trends, and approving high-value operational requests (inventory adjustments, cancellations, and financial vouchers).
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2. Super Admin Navigation Map & Access Scope

Super Admins possess **Company-Wide Scope**, giving you unrestricted visibility across all physical stations (HQ, Branches, Regional Outlets) and all 14 system modules.

Top Navigation Areas for Administrators

- Executive Dashboard (/):** Real-time analytics, revenue trends, and attention queues.
 - User & Access Management (/settings/users):** Account creation, role assignments, station scope bindings.
 - Roles & Permissions (/settings/roles):** Granular permission key definitions and custom role creation.
 - Audit & Compliance Center (/audit):** Cryptographic hash-chain verification and audit log exports.
 - Company & Business Units (/settings):** Legal branding, logo management, payment gateways, and business unit structures.
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3. Section 1: System Onboarding & User Administration

3.1 Creating New User Accounts

To onboard a new employee or manager onto the platform:

1. Navigate to **System Settings** → **User Management**.
2. Click **Add New User**.
3. Fill in the required user details:
 - **Username**: Unique login identifier (e.g., **john.doe**).
 - **Email Address**: Official company email (e.g., **j.doe@aauchamo.local**).
 - **Full Name**: First Name and Last Name.
 - **Initial Password**: Set a strong temporary password meeting security requirements (min 8 characters, uppercase, number, symbol).
4. Select the **Primary Role** (e.g., *Operations Manager, Finance Officer, Sales Officer*).
5. Assign **Station Operating Scopes**:
 - For location-bound staff: Select their specific assigned station (e.g., **Kano Station**).
 - For company-wide staff: Check **Company-Wide Access** (enables visibility across all stations).
6. Click **Save & Activate User Account**.

3.2 Linking User Accounts to Physical Staff Records

To enable staff members to use the **Mobile PWA Attendance Punch Clock App**, their system user account must be linked to a physical **Staff Employee Record**:

1. Navigate to **HR & Staff Management** → **Staff Catalogue**.
2. Edit or create the employee's physical profile.
3. In the **System User Account** dropdown, select the created user login (**john.doe**).
4. Click **Save Staff Profile**. The employee can now use their mobile phone to clock in/out.

3.3 Deactivating or Restricting User Accounts

If an employee leaves the company or changes operational roles:

1. Navigate to **User Management** and locate the user profile.
2. Click **Change Status** → **Deactivate Account**.
3. Deactivation immediately revokes all active JWT session tokens. The user will be logged out automatically and blocked from future logins.

4. Section 2: Security Controls & Session Management

4.1 Automated Account Locking Policy

To protect against brute-force password guessing attacks:

- The system tracks failed login attempts per username and IP address.
- After **5 consecutive failed password attempts**, the account is automatically locked for **15 minutes**.
- As a Super Admin, you can manually unlock a user account before the 15-minute timer expires by clicking **Unlock User** in the User Management panel.

4.2 Managing Active User Sessions

Super Admins can view and manage all active logged-in sessions across the enterprise:

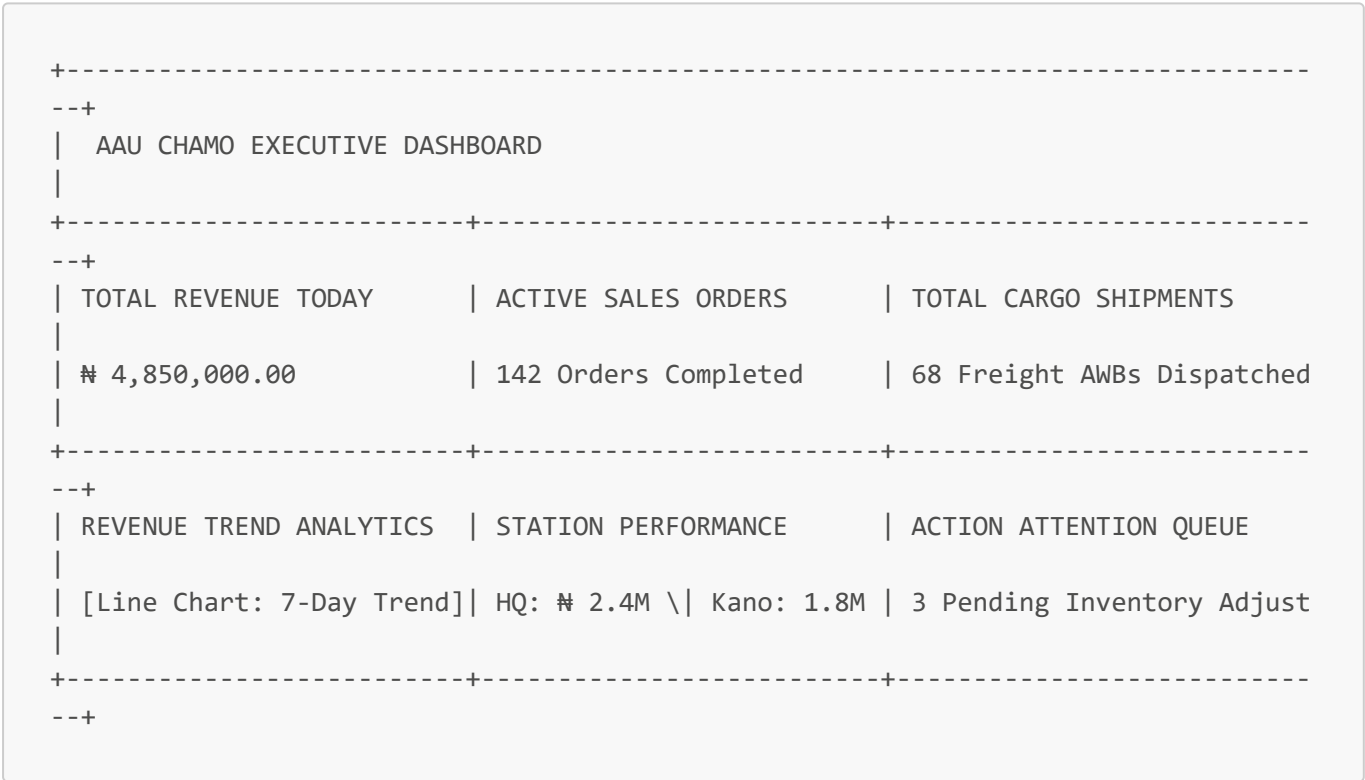
1. Navigate to **Security** → **Active Sessions**.
2. View active user sessions, including:
 - User Name & Role
 - Login Timestamp & Device User-Agent
 - IP Address & Location
3. To force-logout a suspicious session or lost device, click **Revoke Session**.

4.3 Data Encryption Standard

- Sensitive employee national IDs, salary figures, and payment tokens are encrypted at rest using **AES-256-GCM authenticated encryption**.
- The 32-byte encryption key is managed via the environment variable `DATA_ENCRYPTION_KEY`.

5. Section 3: Executive Dashboard Operations (Module 01)

The **Executive Dashboard** is the command center for Super Admins to monitor real-time company performance.



5.1 Real-Time Executive KPIs

- **Total Revenue (Today & MTD):** Consolidated sales across POS counters, cargo freight, and ticket bookings.
- **Active Sales Volume:** Total transactions completed today.
- **Cargo Freight Volume:** Total AWBs booked and dispatched.
- **Pending Approvals Count:** High-priority tasks awaiting management decision.

5.2 Multi-Station Performance Comparison

Super Admins can filter dashboard analytics by:

- **All Stations (Consolidated)**: Company-wide totals.
- **Individual Station Scope**: Filter metrics specifically for HQ, Kano, Abuja, or regional stations to compare productivity and revenue contribution.

5.3 Action Attention Queue Management

The Attention Queue alerts Super Admins to items needing authorization:

1. **Maker-Checker Inventory Adjustments**: Stock count deltas submitted by station managers.
2. **High-Value Expense Vouchers**: Financial payout requests exceeding station limits.
3. **Cargo Shipment Correction Requests**: Modifications requested after AWB dispatch.

6. Section 4: Audit Trail & Integrity Verification

AAU Chamo ERP implements a **Cryptographic Hash-Chain Audit System**. Every transaction, stock movement, user login, and price change creates an immutable **AuditEvent** entry.

6.1 Inspecting Audit Logs

1. Navigate to **Compliance** → **Audit Trail**.
2. Filter audit events by:
 - **Actor**: Specific employee who performed the action.
 - **Action Type**: E.g., `sales.create`, `inventory.adjust`, `auth.login_failed`.
 - **Date Range** & **Station ID**.
3. Click any audit entry to view the **Before** and **After** state snapshots.

6.2 Verifying Audit Chain Cryptographic Integrity

To verify that no database record has been altered or deleted directly by unauthorized database access:

1. Navigate to **Audit Trail** → **Verification**.
2. Click **Run Audit Chain Integrity Verification**.
3. The system recalculates SHA-256 hash digests sequentially from the genesis block to the current record.
4. **Verification Outcome**:
 - **PASS (Green)**: The audit chain is unbroken and 100% tamper-free.
 - **ALERT (Red)**: Indicates a hash mismatch or broken chain sequence, pin-pointing the exact record ID affected.

6.3 Exporting Audit Reports

Click **Export Audit Log (CSV)** to generate an officially timestamped audit report for board reviews or external financial auditors.

7. Section 5: Step-by-Step Daily Super Admin Operating Procedures

Morning Operational Readiness Routine (08:00 AM)

- 1. Log in to the ERP using your Super Admin credentials.
- 2. Check the **Executive Dashboard**:
 - Verify all physical stations have logged in and opened POS cash sessions.
 - Review the **Attention Queue** for overnight notifications or pending approvals.
- 3. Check the **Attendance Portal** (</api/staff/attendance>) to ensure key station managers have clocked in via their mobile PWA app.

Mid-Day Operational Oversight (01:00 PM)

- 1. Review **Station Performance Cards** to identify any low-performing stations or inventory shortages.
- 2. Process pending **Maker-Checker Requests** (inventory adjustments or credit limit increases).
- 3. Monitor **Active Sessions** to ensure remote users are operating securely.

End-of-Day Closing & Governance (06:00 PM)

- 1. Verify that all station POS cash sessions have been closed and reconciled with 0 cash variance.
- 2. Run **Audit Chain Verification** to confirm database integrity.
- 3. Review consolidated daily sales summary reports.

8. Section 6: Emergency Administration & Troubleshooting Guide

Issue / Scenario	Root Cause	Admin Solution Step
User locked out ("Too many failed attempts")	5 invalid password attempts	Go to User Management → Select User → Click Unlock Account .
POS Cash Drawer cannot close	Unclosed pending sales orders	Navigate to Sales Management → Filter Pending → Cancel or Complete Open Drafts .
Mobile Punch Clock GPS Error on Phone	Browser location permission denied	Guide user to phone settings: Settings → Safari/Chrome → Permissions → Location → Allow .
Stale Version Conflict Error (409)	Simultaneous editing by 2 managers	Refresh the record view to load the latest optimistic locking version before editing.